CAMARINES SUR POLYTECHNIC COLLEGES
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COLLEGE *of* COMPUTER STUDIES

Requirements Document

Faculty Consultation Booking System

Midterm Project - Campus System, Round 1

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GROUP 6

Project Overview

The Faculty Consultation Booking System is a web application that lets students request consultation time with faculty members, and lets faculty publish their availability and manage incoming requests. The system replaces informal, ad hoc scheduling (walk-ins, messaging apps, posted office hours) with a single, role-based platform that tracks availability and booking status from request to completion.

The problem being addressed is that students have no reliable way to see when a faculty member is available, and faculty have no structured way to receive, approve, or track consultation requests. This leads to missed meetings, double-booked time, and no record of whether a request was ever addressed.

Users

People who directly interact with the system through its interface.

Role	Description
Student	Registers with an @my.cspc.edu.ph email, views faculty from their own department and their published availability, requests a consultation (date/time, duration, topic, notes), and tracks, cancels or reschedules their own bookings.
Faculty	Registers with an @my.cspc.edu.ph email, publishes and activates/deactivates availability slots, and reviews consultation requests from students: approves or rejects pending requests and marks completed consultations.
Admin	Oversees the system: views summary counts, all registered users and recent bookings, and activates or deactivates user accounts. Admin accounts are not created through registration.

Stakeholders

People or groups who are affected by, or have an interest in, the system, even if they do not use it directly.

Stakeholder	Relationship to the System
Students	Primary users; need a reliable, transparent way to reach faculty for consultation.
Faculty members	Primary users; need an organized way to manage consultation availability and requests without being overwhelmed by informal messages.

Stakeholder	Relationship to the System
Department Chair / Registrar's Office	May rely on consultation records as evidence that academic support is being provided to students.
Course Instructor (Mr. Mamansag)	Evaluates the system as a midterm project deliverable and assesses its functionality, requirements fit, and development process.
Development Team	Responsible for building, testing, maintaining, and defending the system.

Functional Requirements

What the system must do.

ID	Requirement
FR-1	The system shall allow a new user to register as a Student or Faculty by providing a full name, an @my.cspc.edu.ph email (or an approved demo email), a department from the approved list, and a password of at least 8 characters entered twice. The password shall be stored as a bcrypt hash.
FR-2	The system shall reject registration when required fields are missing or invalid, when the passwords do not match, or when the email is already registered. The system shall not allow the Admin role to be chosen at registration.
FR-3	The system shall allow a registered, active user to log in with email and password and shall establish a session. Deactivated accounts shall not be able to log in.
FR-4	The system shall allow a logged-in user to log out, which ends the session.
FR-5	The system shall allow a Faculty user to publish an availability slot with a future start date/time and a later end date/time.
FR-6	The system shall allow a Faculty user to deactivate and reactivate their own availability slots.
FR-7	The system shall allow a Student user to view active faculty members from their own department and the upcoming active availability of the selected faculty member.
FR-8	The system shall allow a Student user to request a consultation by choosing a faculty member, a future date/time, a duration (15 to 120 minutes in 15-minute steps), a topic (1 to 120 characters), and optional notes (up to 1000 characters). A new request shall start with the status "pending".
FR-9	The system shall accept a consultation request only if the requested time falls inside the faculty member's active published availability and does not overlap another pending or approved consultation of that faculty member.
FR-10	The system shall allow a Student user to view all of their own bookings with the faculty name, topic, time, duration, and status (pending, approved, rejected, cancelled, or completed).
FR-11	The system shall allow a Student user to cancel their own pending or approved booking, provided it has not started yet.
FR-12	The system shall allow a Student user to reschedule their own pending or approved future booking to a new valid time. The new time shall pass the same availability and conflict checks, and the booking shall return to "pending".

ID	Requirement
FR-13	The system shall allow a Faculty user to view all bookings made with them, including the student's name, email, topic, and notes.
FR-14	The system shall allow a Faculty user to approve or reject a pending booking, and to mark an approved booking as completed only after its start time has passed. Other status changes shall be refused.
FR-15	The system shall allow an Admin user to view summary counts (active students, active faculty, total bookings, pending bookings), the list of all registered users, and the most recent bookings.
FR-16	The system shall allow an Admin user to activate or deactivate any user account except their own.
FR-17	The system shall restrict every protected page by role (Student, Faculty, or Admin). A visitor who is not logged in shall be redirected to the login page, and a logged-in user with the wrong role shall see an "Access denied" page, including when the URL is typed directly.
FR-18	The system shall ensure that students can change only their own bookings and faculty can change only bookings assigned to them and their own availability slots.

Non-Functional Requirements

How the system should perform.

ID	Requirement
NFR-1	Security: Passwords shall be hashed with bcrypt before storage; plain-text passwords shall never be stored or logged. Authentication state shall be kept in server-side sessions (stored in MySQL, with the session ID regenerated at login) and not exposed in the URL. Security headers shall be set with Helmet.
NFR-2	Data integrity: All form input shall be validated on the server (required fields, lengths, email format and domain, allowed roles and departments, future dates, valid durations, schedule conflicts) and stored using parameterized SQL queries to prevent SQL injection.
NFR-3	Access control: Role checks shall be enforced by server-side middleware on every protected route, not only by hiding links in the interface.
NFR-4	Persistence: Accounts, availability slots, and bookings shall be stored in a MySQL database so they remain after page refreshes and server restarts.
NFR-5	Usability: The interface shall use plain, consistent navigation and clear success and error messages so that a first-time user can register, log in, and complete their main task without outside instructions.
NFR-6	Availability: The application shall run reliably on a local XAMPP (MySQL) and Node.js environment during a live demonstration without needing a restart.
NFR-7	Maintainability: The code shall be organized into separate parts (routes per role, middleware, shared utilities, database configuration, and views) so individual features can be changed independently.
NFR-8	Configuration: Secrets such as the session secret and database password shall be kept in a .env file that is not committed to the repository; a .env.example file shall document the required values.

Use Cases

Each use case describes a major system function in terms of the actor involved, the precondition that must be true before it can occur, the steps taken, and the expected result.

UC-1: Register an Account

Field	Description
Actor	Student or Faculty (unregistered visitor)
Precondition	The visitor does not have an account and is on the login page.
Steps	1. The visitor opens the Register page. 2. The visitor enters full name, @my.cspc.edu.ph email, department, role (Student or Faculty), and a password twice. 3. The visitor submits the form. 4. The system validates the input and checks that the email is not already registered. 5. The system hashes the password and stores the account.
Expected Result	A new account is created and the visitor is sent to the login page with a success message. If any check fails (invalid input, mismatched passwords, duplicate email), an error message is shown and no account is created.

UC-2: Log In

Field	Description
Actor	Registered user (Student, Faculty, or Admin)
Precondition	The user has an active account and is on the login page.
Steps	1. The user enters email and password. 2. The system finds the account and compares the password with the stored hash. 3. If valid and the account is active, the system creates a session with the user's ID, name, email, role, and department.
Expected Result	The user is redirected to the page for their role (Student: My consultations, Faculty: Faculty dashboard, Admin: Admin dashboard). If the credentials are wrong or the account is deactivated, a generic error is shown and no session is created.

UC-3: Publish Availability

Field	Description
Actor	Faculty
Precondition	The Faculty user is logged in.
Steps	1. The Faculty user opens the Faculty dashboard. 2. The user enters a future start date/time and a later end date/time. 3. The user submits the form. 4. The system validates the times and stores the slot linked to the Faculty user, marked active.
Expected Result	The slot appears on the faculty dashboard and becomes available to students of the same department. The Faculty user can later deactivate or reactivate the slot. A user without the Faculty role is denied access.

UC-4: Request a Consultation

Field	Description
Actor	Student
Precondition	The Student user is logged in and at least one faculty member in their department has active availability.
Steps	1. The Student opens My consultations and selects a faculty member to see their availability. 2. The Student enters a date/time, duration, topic, and optional notes. 3. The Student submits the form. 4. The system checks the input, confirms the time is inside the faculty member's availability, and confirms it does not overlap another pending or approved booking.
Expected Result	A booking with status "pending" is created and shown in the student's list. If a check fails, an error message explains why and no booking is created.

UC-5: Manage a Booking Request

Field	Description
Actor	Faculty
Precondition	The Faculty user is logged in and has at least one booking.
Steps	1. The Faculty user opens the Faculty dashboard and reviews the booking's student, topic, notes, and time. 2. For a pending booking, the user clicks Approve or Reject. 3. After the consultation time has passed, the user may mark an approved booking as Completed.
Expected Result	The booking status is updated and the new status is visible to the student on their page. A change that is not allowed (for example completing a booking that has not started) is refused with an error message.

UC-6: Cancel a Booking

Field	Description
Actor	Student
Precondition	The Student is logged in and has a pending or approved booking that has not started.
Steps	1. The Student opens My consultations. 2. The Student clicks Cancel on the booking.
Expected Result	The status becomes "cancelled" and the time can be booked again. A booking that already started, or that belongs to another student, cannot be cancelled.

UC-7: Reschedule a Booking

Field	Description
Actor	Student
Precondition	The Student is logged in and has a pending or approved booking that has not started.
Steps	1. The Student chooses Reschedule on the booking. 2. The Student enters a new date/time and duration. 3. The system checks availability and conflicts.

Field	Description
Expected Result	The booking time is updated and its status returns to "pending" for the faculty member to approve again. If the new time is unavailable or conflicts with another consultation, an error is shown and the booking is unchanged.

UC-8: Monitor Users and Bookings

Field	Description
Actor	Admin
Precondition	The Admin user is logged in.
Steps	1. The Admin opens the Admin dashboard. 2. The system shows summary counts, all users, and recent bookings. 3. The Admin may activate or deactivate a user account.
Expected Result	The Admin sees every registered user and the most recent bookings across the system. A deactivated user can no longer log in. The Admin cannot deactivate their own account. Student and Faculty users are denied access to this page.

Access Control Summary

Role-based access is enforced through server-side middleware rather than by hiding links in the interface, so a user cannot bypass it by typing a URL directly.

Function	Student	Faculty	Admin
Register / Log in	Allowed	Allowed	Allowed
Publish and manage availability	Denied	Allowed	Denied
Request, cancel, reschedule consultations	Allowed	Denied	Denied
Approve, reject, complete bookings	Denied	Allowed	Denied
View all users and recent bookings	Denied	Denied	Allowed
Activate / deactivate accounts	Denied	Denied	Allowed